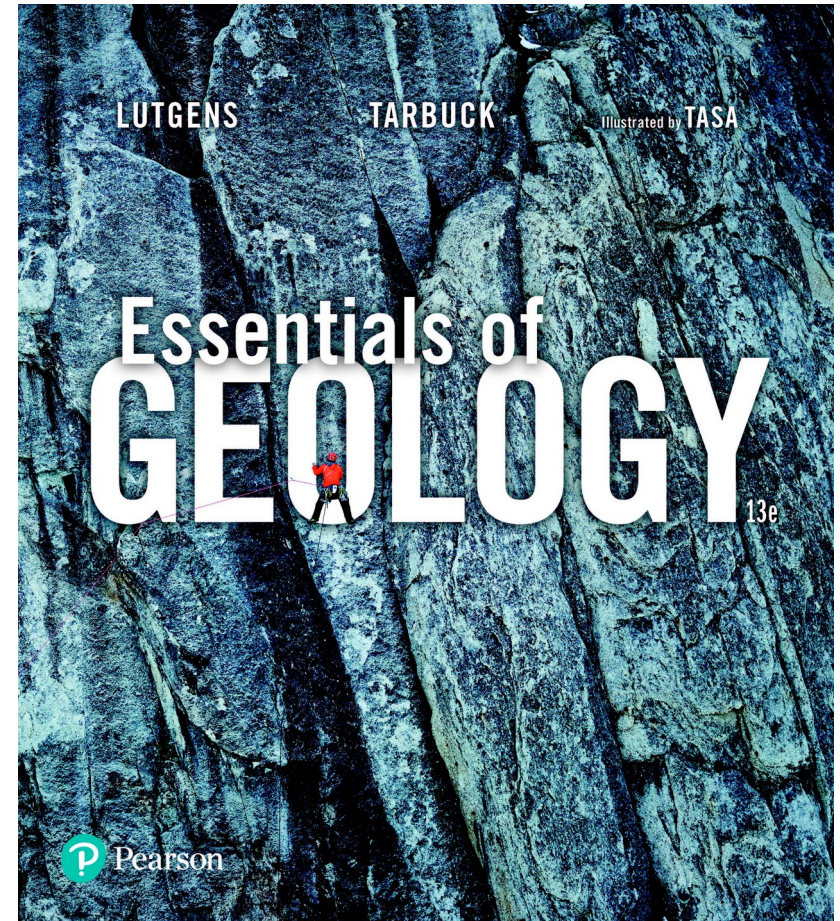


Essentials of Geology

Thirteenth Edition

Crustal Deformation & Mountain Building

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Scottsdale Community College



Learning Objectives

- **LO 11.1** Describe the three types of differential stress and identify the tectonic setting most commonly associated with each. Differentiate stress from strain and brittle from ductile deformation.
- **LO 11.2** List and describe five types of folds.
- **LO 11.3** Sketch and briefly describe the relative motion of rock bodies located on opposite sides of normal, reverse, and thrust faults as well as both types of strike-slip faults.
- **LO 11.4** Locate and name Earth's major mountain belts on a world map.

Learning Objectives (cont.)

- **LO 11.5** Sketch a cross section of an Andean-type mountain belt and describe how its major features are generated.
- **LO 11.6** Summarize the stages in the development of an Alpine-type mountain belt such as the Appalachians.
- **LO 11.7** Explain the principle of isostasy and how it contributes to the elevated topography of mountain belts.

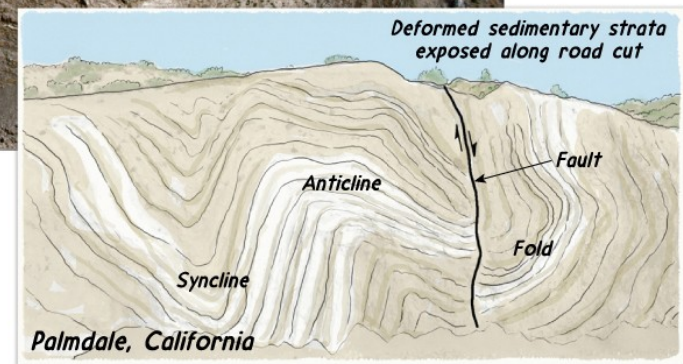
11.1 Crustal Deformation

- Deformation is a general term that refers to all changes in the shape or position of a rock body in response to stress
- Rock or geologic structures are the features that result from forces generated by the interactions of tectonic plates
 - Includes folds, faults, and joints

11.1 Crustal Deformation (cont.)

Tutorial

- Deformed Sedimentary Strata



Geologist's Sketch

11.1 Crustal Deformation (cont.)

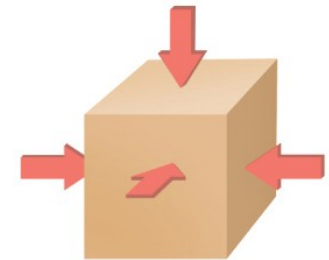
- Stress: the force that deforms rocks
 - Stress is the force that deforms rocks
 - When stresses acting on a rock exceed its strength, the rock will deform by flowing, folding, fracturing, or faulting.
 - The magnitude is a function of the amount of force applied to a given area.

11.1 Crustal Deformation (cont.)

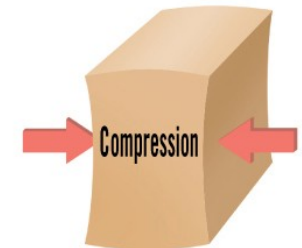
- Stress: The force that deforms rocks
 - Stress applied uniformly in all directions is confining pressure, reducing volume.
 - Stress applied unequally in different directions is called differential stress, causing most deformation.

11.1 Crustal Deformation (cont.)

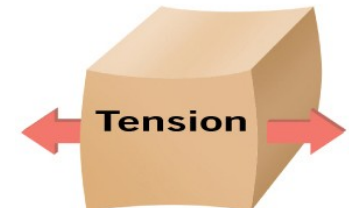
- **Stress:** The force that deforms rocks
 - Types of stress
 - Compressional stress squeezes a rock and shortens a rock body.
 - Tensional stress pulls apart a rock unit and lengthens it.
 - Shear stress produces a motion similar to slippage that occurs between individual playing cards when the top of the stack is moved relative to the bottom.



A. Confining pressure



B. Compressional stress (shortening)



C. Tensional stress (stretching)

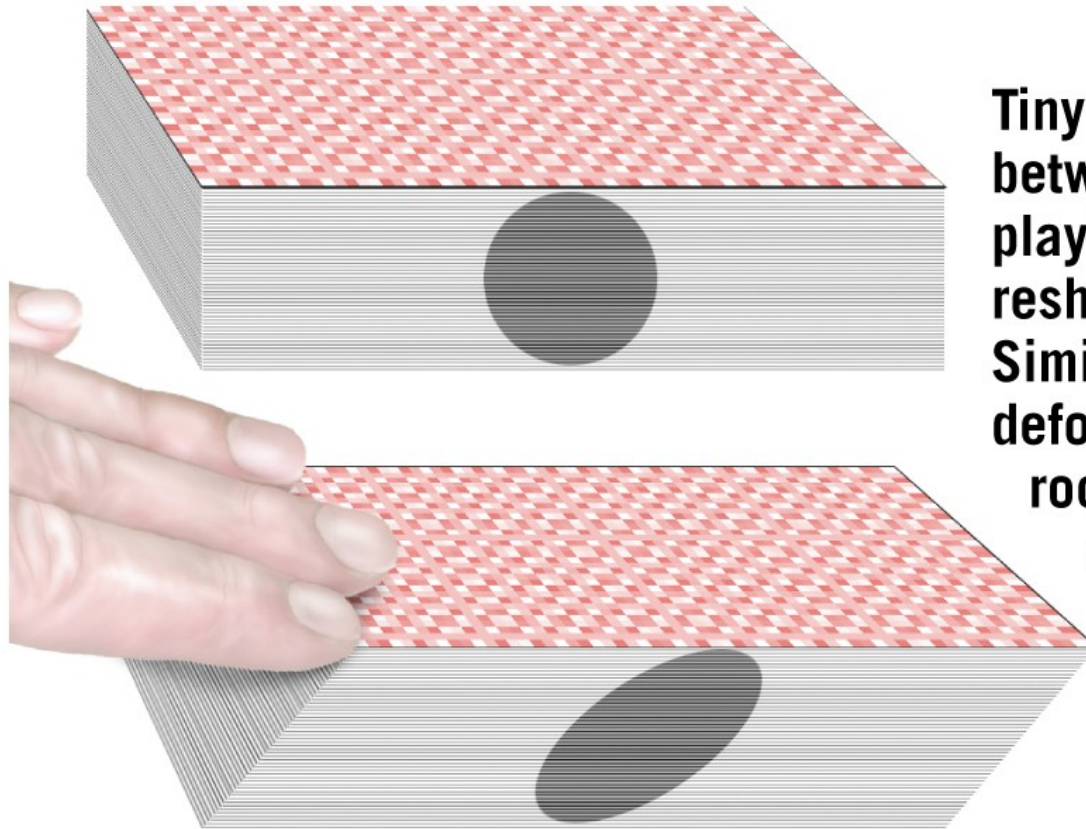


D. Shear stress (sliding and tearing)

11.1 Crustal Deformation (cont.)

- Strain: A change in shape caused by stress
 - Strain is the change in shape of a rock caused by differential stress.
 - Strained bodies lose their original configuration during deformation.

11.1 Crustal Deformation (cont.)



Tiny movements between adjacent playing cards can reshape the deck. Similarly, *ductile* deformation in a rock body may be accomplished through many small *brittle* offsets.

11.1 Crustal Deformation (cont.)

- Elastic, brittle, and ductile deformation
 - Elastic deformation: The rock returns to nearly its original size and shape when the stress is removed.
 - Once the elastic limit (strength) of a rock is surpassed, it either bends (ductile deformation) or breaks (brittle deformation).

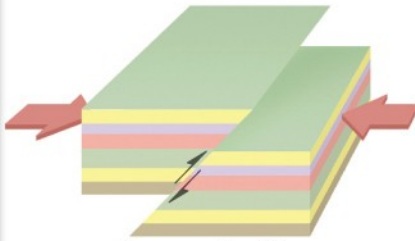
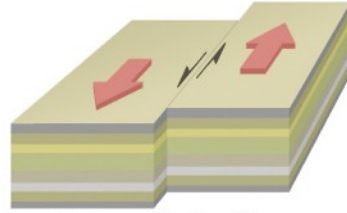
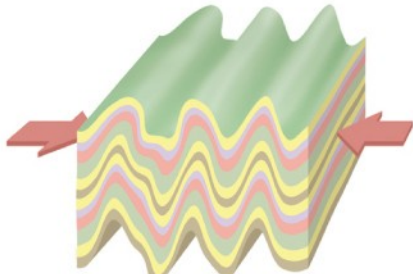
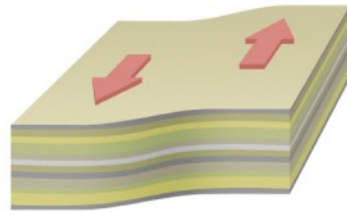
11.1 Crustal Deformation (cont.)

- Factors that affect rock strength
 - Temperature: Higher temperature rocks deform by ductile deformation, whereas cooler rocks deform by brittle deformation.
 - Confining pressure: Confining pressure squeezes rocks, making them stronger and harder to break.

11.1 Crustal Deformation (cont.)

- Factors that affect rock strength
 - Rock type: Crystalline igneous rocks generally experience brittle deformation, whereas sedimentary and metamorphic rocks with zones of weakness generally experience ductile deformation.
 - Time: Forces applied over a long period of time generally result in ductile deformation.

Deformation Caused By Three Types of Stress

How Rocks Respond to Differential Stress				
DEPTH		COMPRESSION (Causes shortening)	TENSION (Causes lengthening)	SHEAR (Causes tearing or bending)
SHALLOW	At shallow depths rocks tend to exhibit brittle fracture	 Reverse faulting	 Normal faulting	 Strike-slip faulting
		 Folding	 Stretching	 Shearing
DEEP	At greater depths rocks tend to deform by ductile flow			

11.2 Folds: Rock Structures Formed by Ductile Deformation

- During crustal deformation, rocks are often bent into a series of wavelike undulations called *folds*.
- Characteristics of folds
 - Most folds result from compressional stresses that result in a shortening and thickening of the crust.

11.2 Folds: Rock Structures Formed by Ductile Deformation (cont.)

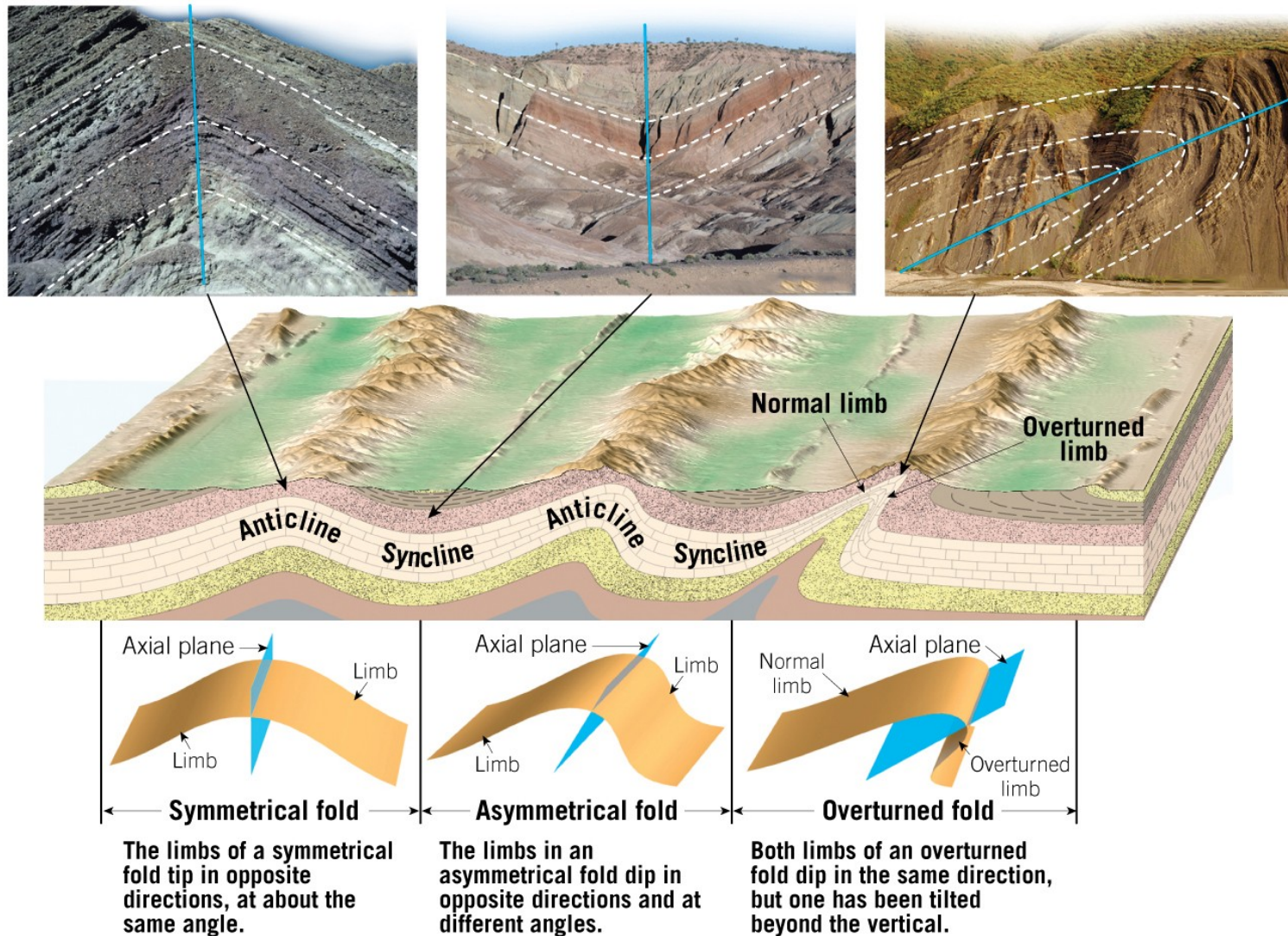
- Anticline and Synclines
 - Anticlines are upfolded or arched sedimentary layers.
 - Oldest strata are in the center.
 - Synclines are downfolded or troughs of rock layers.
 - Youngest strata are in the center.

11.2 Folds: Rock Structures Formed by Ductile Deformation (cont.)

- Anticline and Synclines
 - Depending on their orientation, anticlines and synclines can be described as:
 - Symmetrical: The limbs of the fold are mirror images of each other.
 - Asymmetrical: The limbs of the fold are not identical.
 - Overturned (recumbent): One or both limbs are tilted beyond vertical.
 - Plunging: The axis of the fold penetrates the ground.

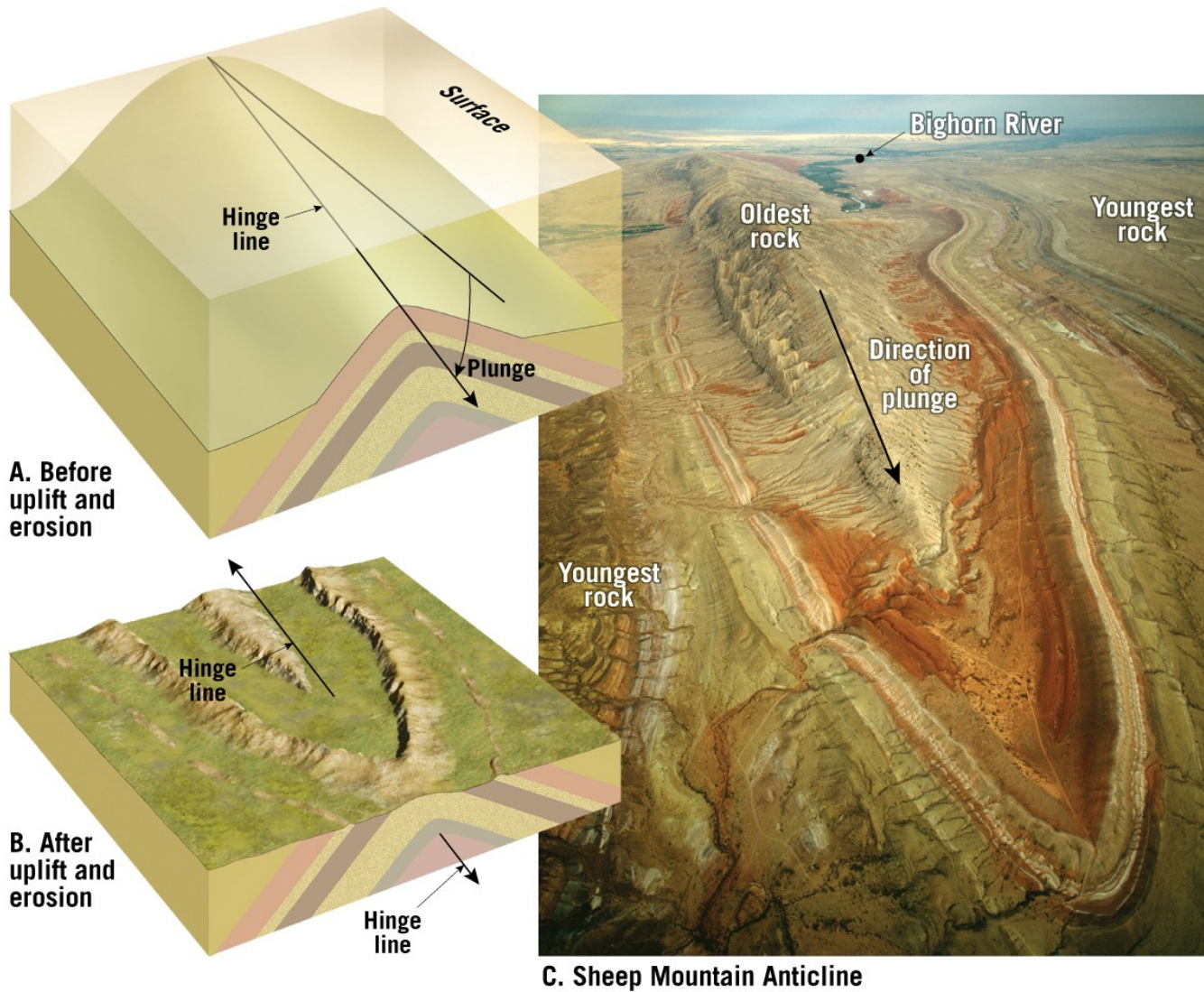
Common Types of Folds

Tutorial



Outcrop Patterns of Plunging Folds

Mobile Field Trip

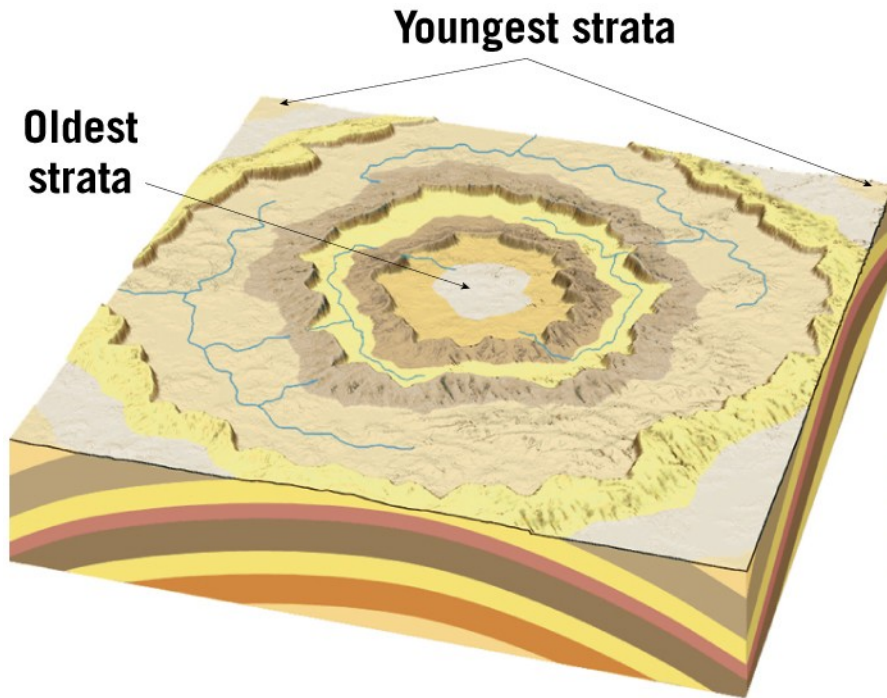


11.2 Folds: Rock Structures Formed by Ductile Deformation

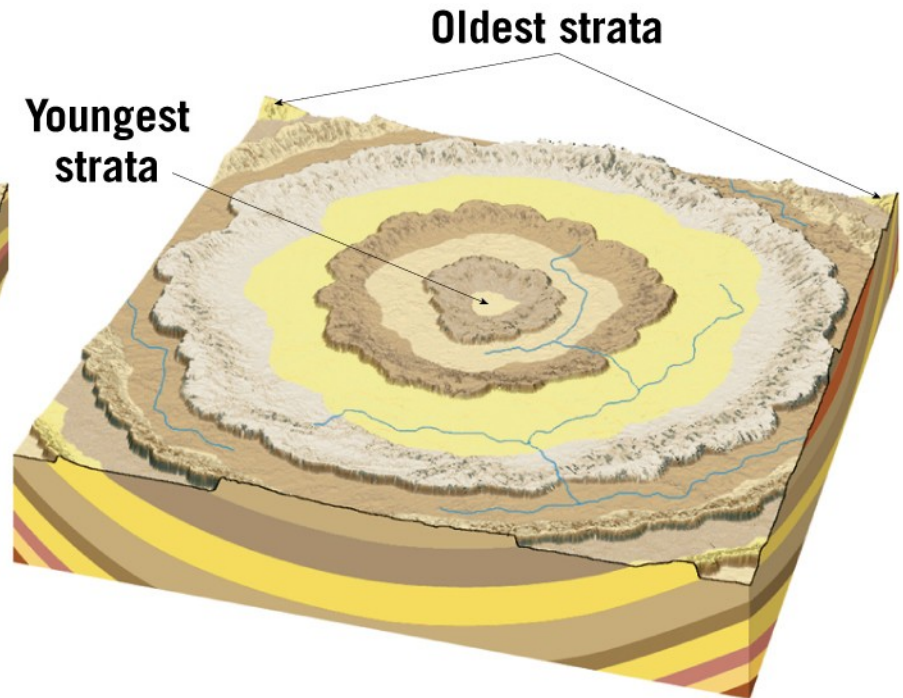
- Domes and basins
 - Domes are upwarped circular features.
 - Oldest rocks are in the center.
 - Basins are downwarped circular features.
 - Youngest rocks are in the center.

Domes versus Basins

Tutorial

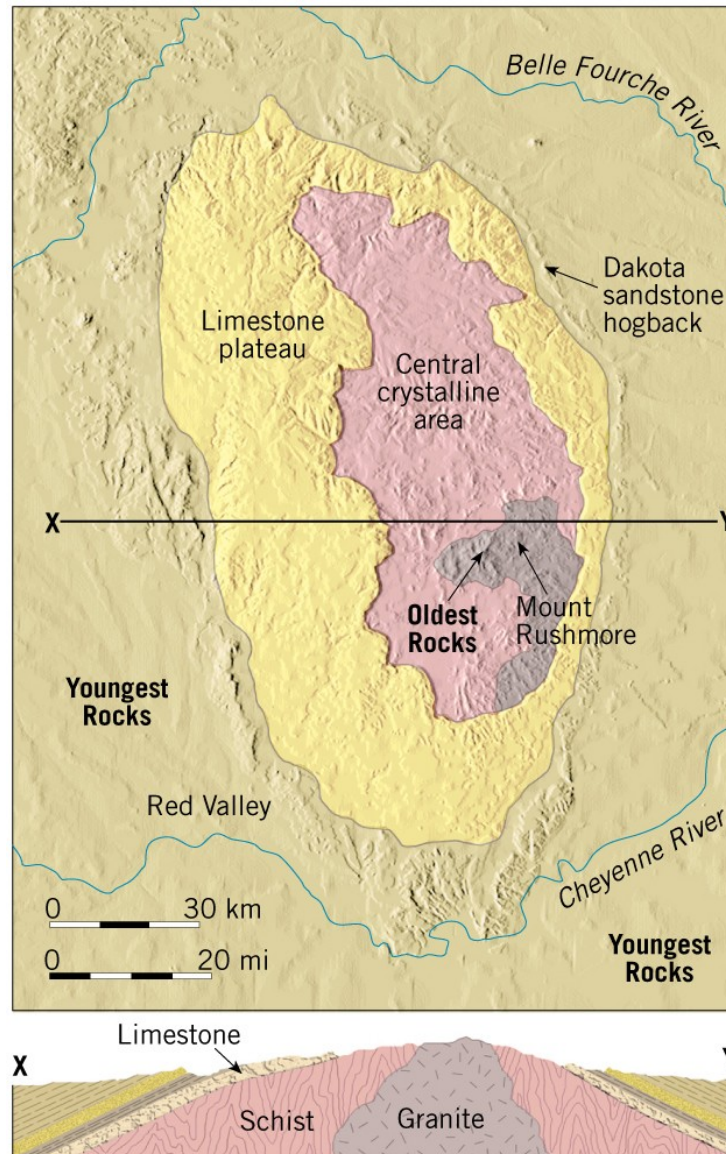


A. Upwarping produces a *dome*.

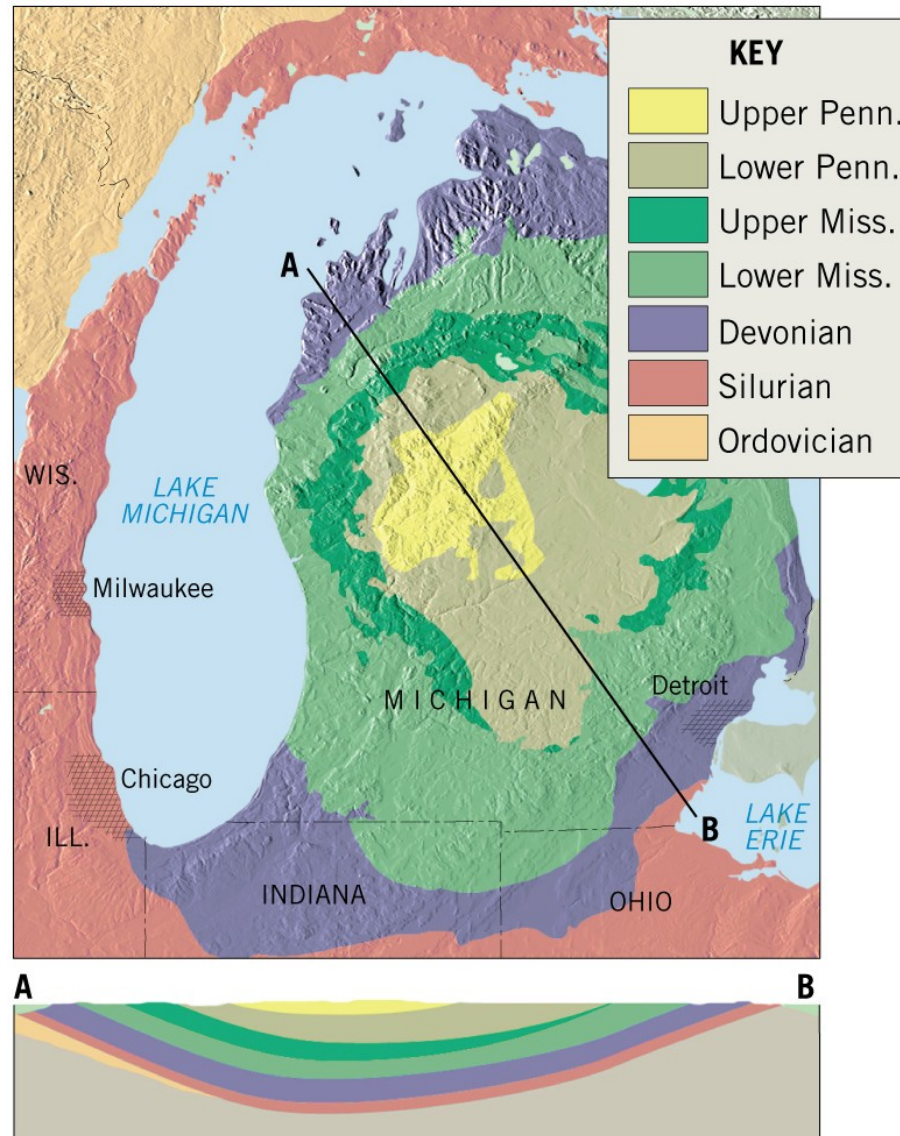


B. Downwarping produces a *basin*.

The Black Hills of South Dakota



The Bedrock Geology of the Michigan Basin

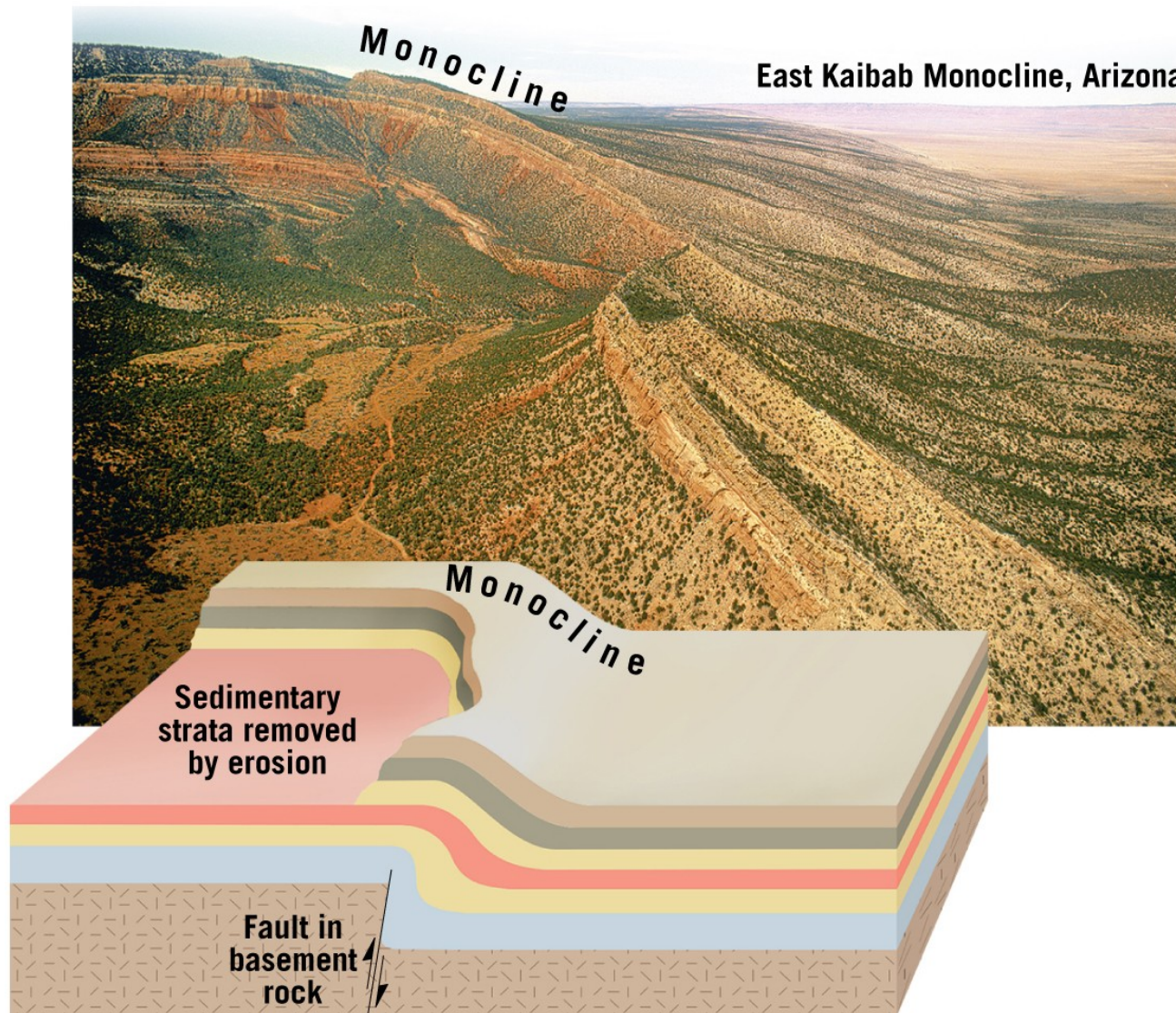


11.2 Folds: Rock Structures Formed by Ductile Deformation

- Monoclines
 - Monoclines are large, steplike folds in otherwise horizontal sedimentary strata.
 - As blocks of basement rocks are displaced upward, the ductile sedimentary strata drape over them.

The East Kaibab Monocline, Arizona

[Condor Video](#)

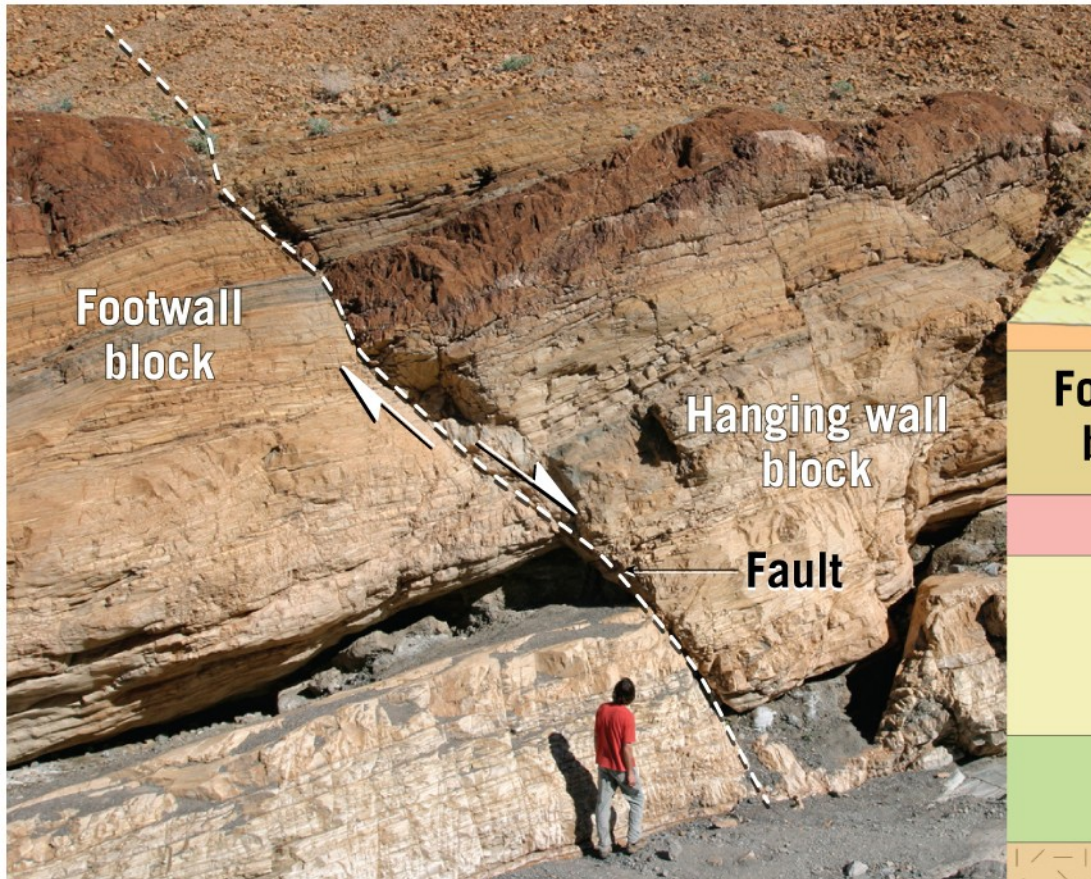


11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

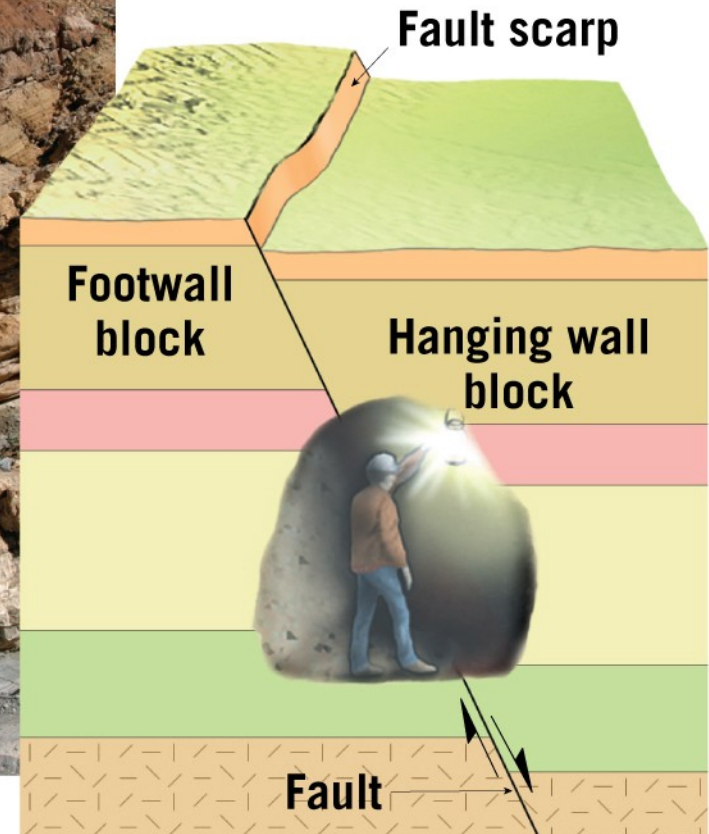
- Joints are fractures in rocks.
- Faults are fractures in rocks, along which motion has occurred.
- Sudden movements along faults are the cause of most earthquakes.

Hanging Wall Block and Footwall Block

[Animation](#)



A.



B.

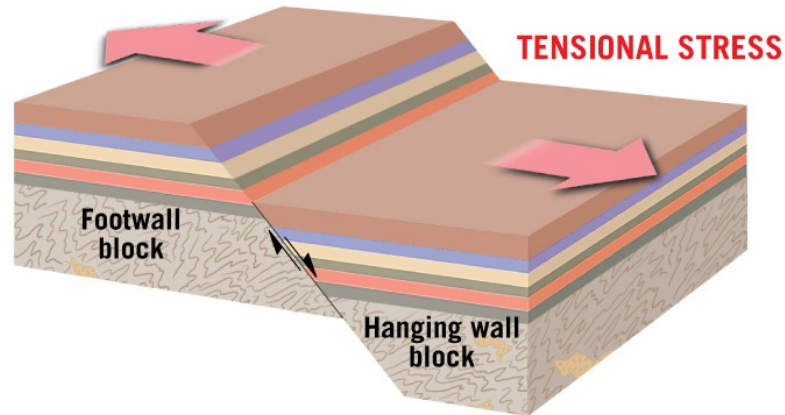
11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

- Dip-slip faults
 - Dip-slip faults occur when movement is parallel to the inclination.
 - The hanging wall is rock surface above the fault.
 - The footwall is the rock surface below the fault.
 - The vertical displacement along the fault produces long, low cliffs called fault scarps.

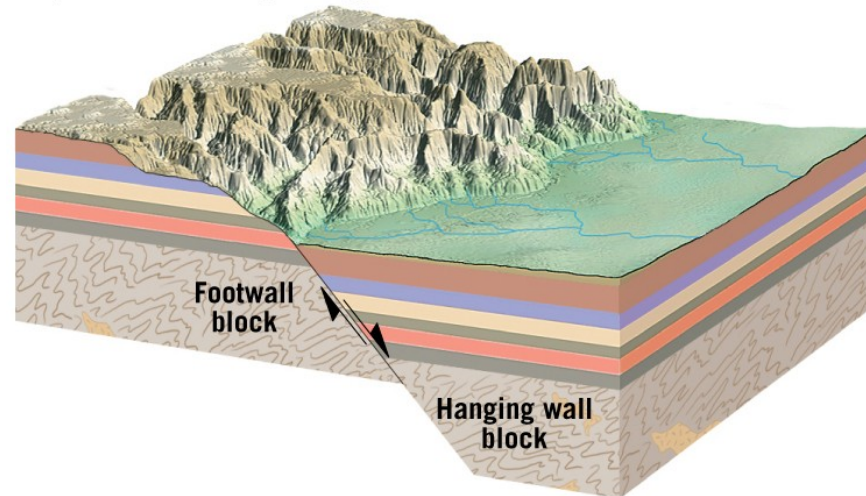
Normal Dip-Slip Fault

Tutorial

A. Idealized block diagram showing a normal fault in which the hanging wall block has moved down relative to the footwall block.



B. Landscape as it might appear after erosion has carved the uplifted block during numerous episodes of displacement along a large normal fault.



11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

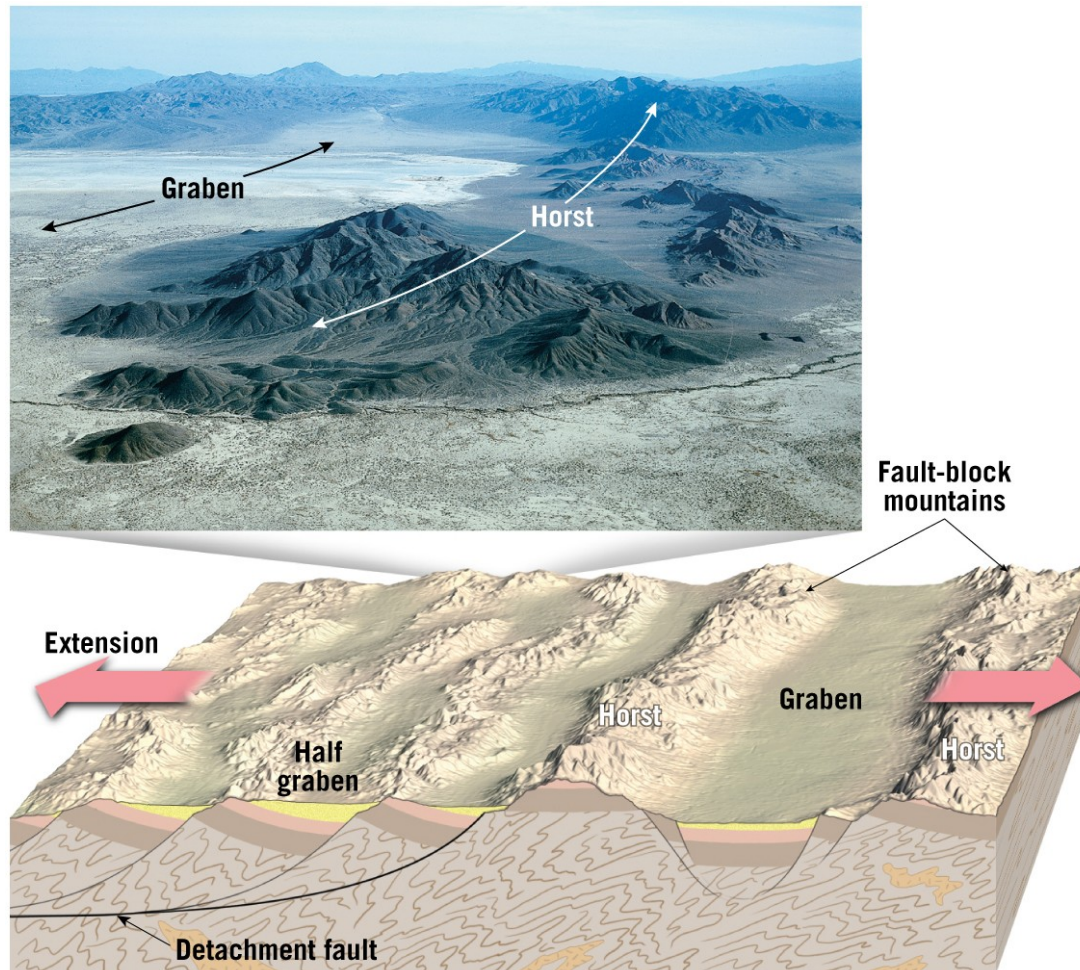
- Dip-slip faults
 - Normal faults are characterized by the hanging wall moving down relative to the footwall.
 - Associated with tensional stress as the rocks pull apart
 - Larger scale normal faults are associated with fault-block mountains.
 - Example: Basin and Range Province
 - Uplifted blocks are called horsts.
 - Down-dropped blocks are called grabens.

11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation (cont.)

- Dip-slip faults
 - Fault-block mountains
 - Half-grabens are tilted fault blocks.
 - Detachment faults represent the boundary between ductile and brittle rock units.

Normal Faulting in the Basin and Range Province

Mobile Field Trip



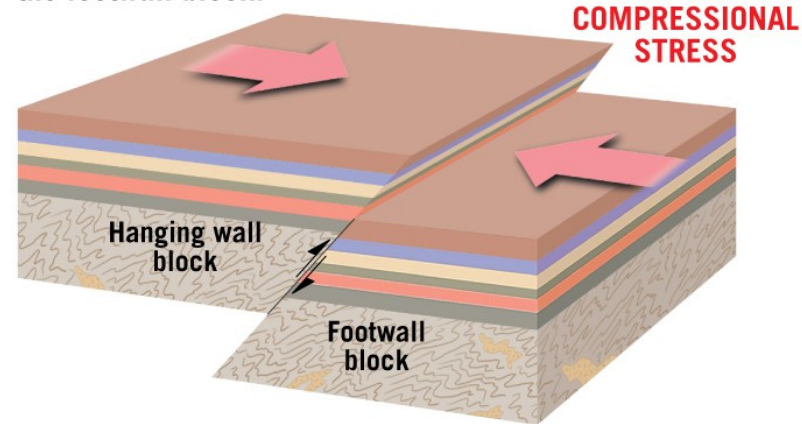
11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

- Dip-slip faults
 - Reverse faults are characterized by the hanging wall moving up relative to the footwall.
 - Associated with compressional stress as the crust shortens.
 - Thrust faults have an angle less than 45 degrees, so the overlying plate moves almost horizontally.
 - Most pronounced along convergent plate boundaries
 - Examples: Alps, Northern Rockies, Himalayas, and Appalachians

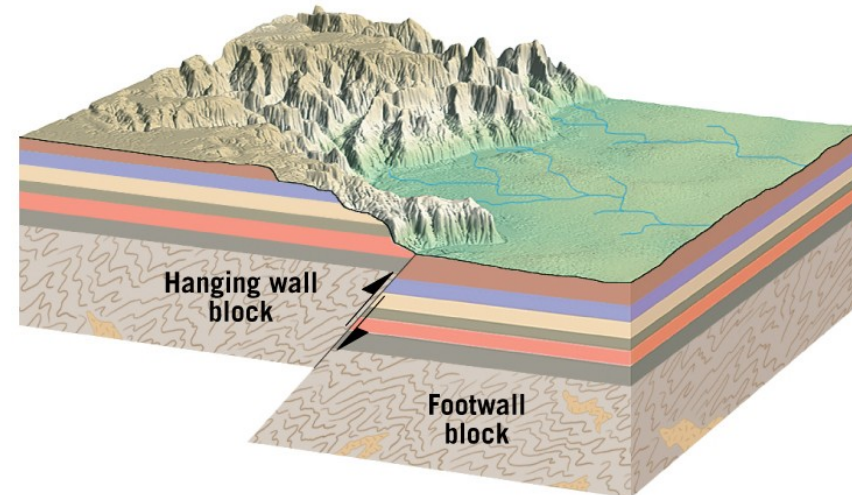
Reverse Faults

[Animation](#)

A. Idealized block diagram showing a reverse fault in which the hanging wall block has moved up relative to the footwall block.



B. Landscape as it might appear after erosion has carved the uplifted block during numerous episodes of displacement along a large reverse fault.



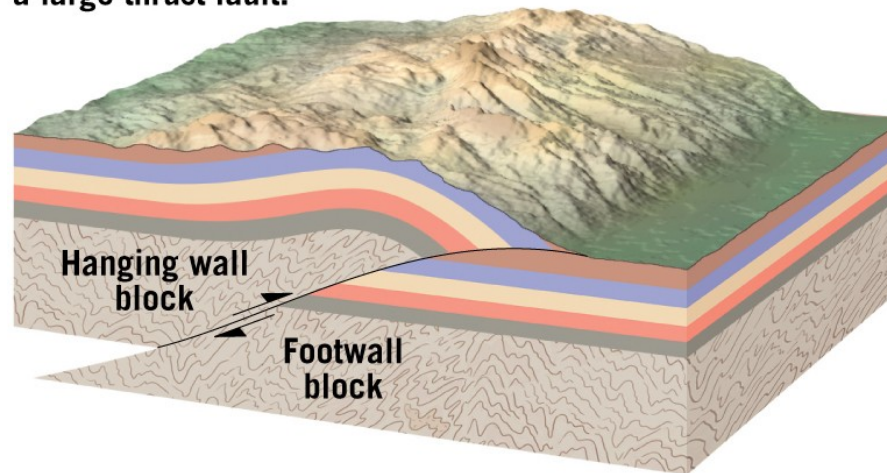
Thrust Fault

[Animation](#)

A. Idealized block diagram showing a thrust fault in which the hanging wall block has moved up relative to the footwall block.



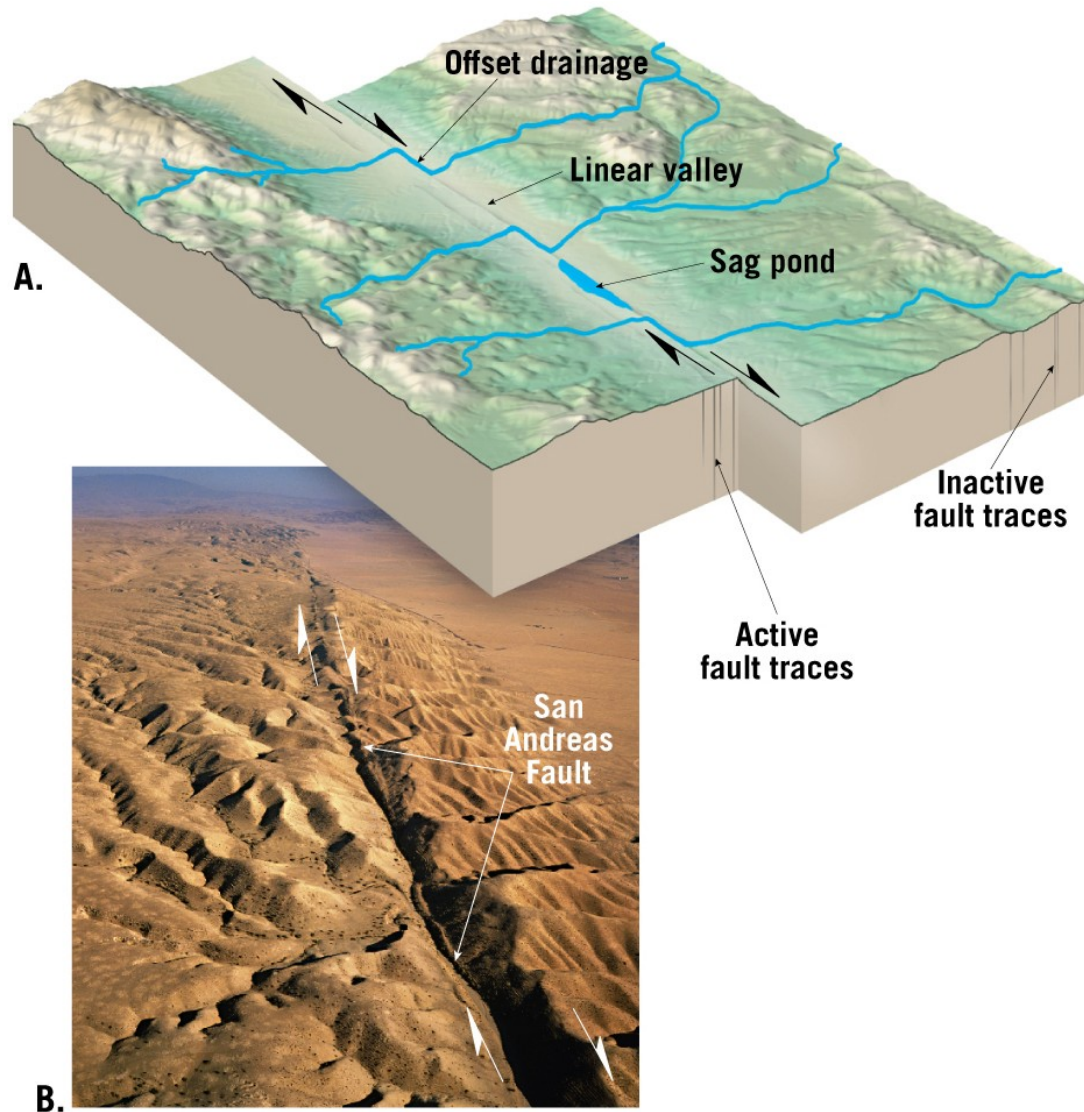
B. Landscape as it might appear after erosion has carved the uplifted block during numerous episodes of displacement along a large thrust fault.



11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

- Strike-slip faults are characterized by placement that is horizontal and parallel to the strike of the fault.
 - Types of strike-slip faults
 - Right-lateral: As you face the fault, the opposite side of the fault moves to the right.
 - Left-lateral: As you face the fault, the opposite side of the fault moves to the left.
- Large strike-slip faults that cut through the crust to accommodate plate motion are called transform faults (e.g., California's San Andreas Fault).

A Strike-Slip Fault



11.3 Faults & Joints: Rock Structures Formed by Brittle Deformation

- Joints are fractures in a rock where there has been no rock movement.
 - Most joints appear in parallel groups.

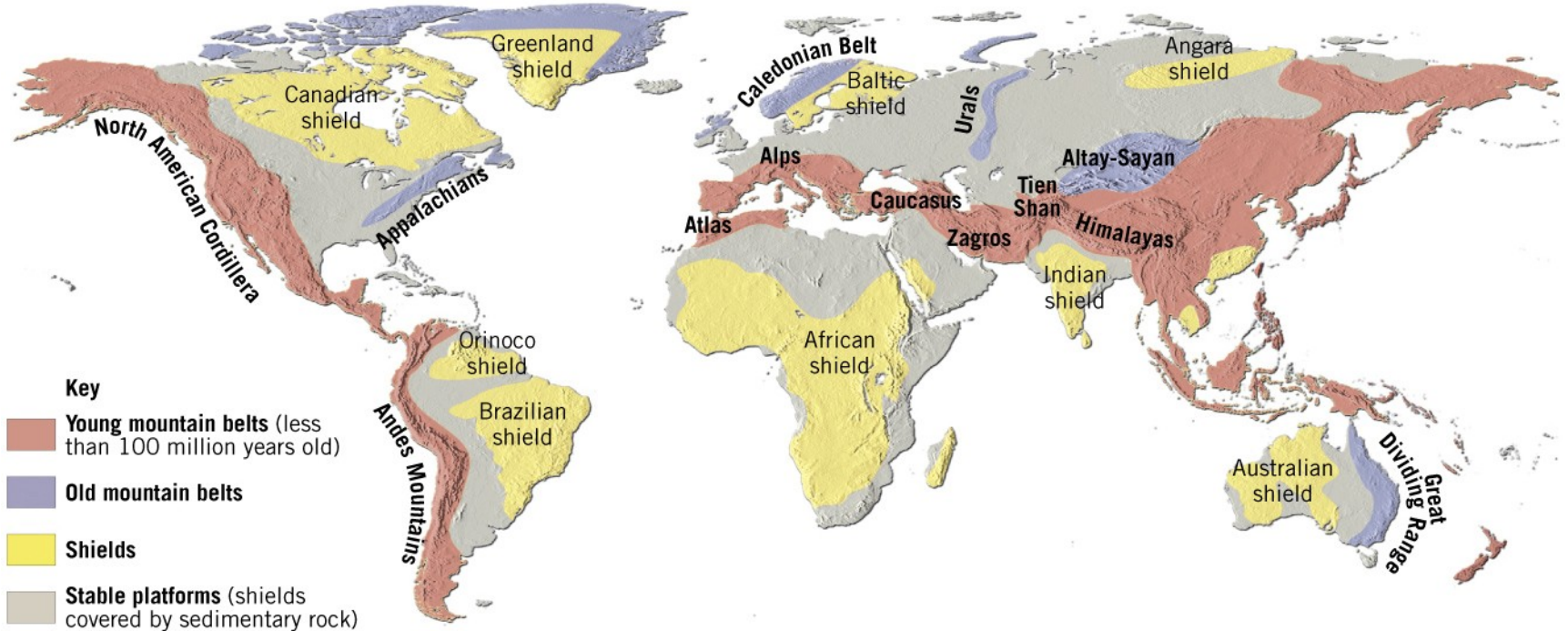
Parallel Joints



11.4 Mountain Building

- Mountain building has occurred during the recent geologic past.
 - American Cordillera
 - The Alpine–Himalaya chain
 - The mountainous terrains of the western Pacific

Earth's Major Mountain Belts



11.4 Mountain Building

- Older Paleozoic-age mountains
 - The Appalachians and the Urals in Russia
 - These mountains are deeply eroded and topographically less prominent.
- Orogenesis is the process that collectively produces a mountain belt.

11.4 Mountain Building (cont.)

- Mountains that display faulted and folded rocks are compressional mountains.
 - Display visual evidence of compressional forces
- Plate tectonics provide a model for orogenesis.
 - Earth's major mountains have formed along convergent plate boundaries.

11.5 Subduction & Mountain Building

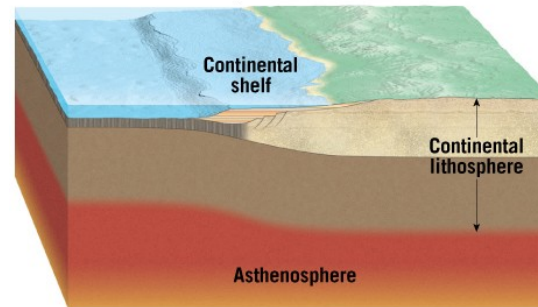
- Island arc–type mountain building
 - Results from the steady subduction of oceanic lithosphere
 - Continued growth can result in the formation of parallel mountainous topography.
 - Consists of igneous and metamorphic rocks

11.5 Subduction & Mountain Building

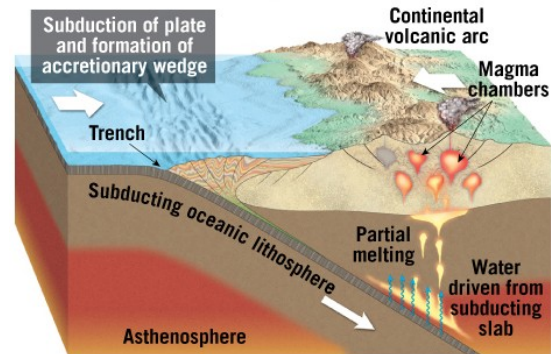
- Andean-type mountain building
 - Subduction beneath a continent rather than oceanic lithosphere
 - Exemplified by the Andes Mountains
 - Starts with a passive continental margin
 - Thick platform of shallow-water sedimentary rocks
 - Eventually, the forces that drive plate tectonics change direction and a subduction zone forms.
 - Oceanic lithosphere must be dense enough to sink.

Andean-Type Mountain Building

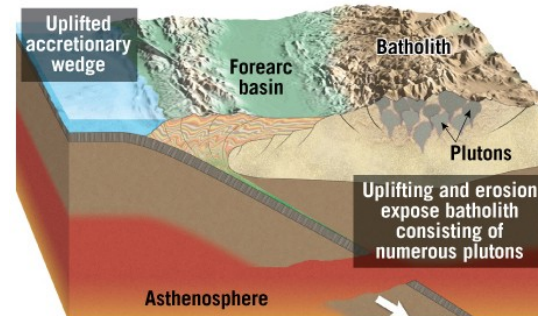
A. Passive continental margin with an extensive platform of sediments and sedimentary rocks.



B. Plate convergence generates a subduction zone, and partial melting produces a continental volcanic arc. Compressional forces and igneous activity further deform and thicken the crust, elevating the mountain belt.



C. Subduction ends and is followed by a period of uplift.



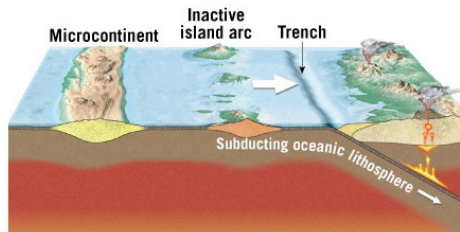
11.6 Collisional Mountain Belts

- Cordilleran-type mountain building
- Associated with the Pacific Ocean (meaning it is unlikely to close)
 - Subduction zones may form island arcs that will eventually collide with a continental crust.
 - The collision and accretion of small slivers of continental crust form the mountainous regions that rim the Pacific.
 - Terranes (crustal fragments of exotic material) make up much of the western United States.

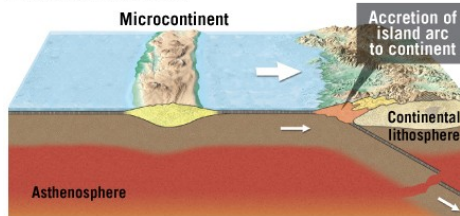
Collision and Accretion of Small Crustal Fragments to a Continental Margin

Tutorial

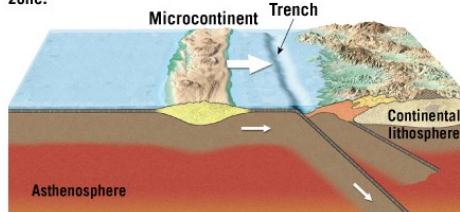
A. A microcontinent and a volcanic island arc are being carried toward a subduction zone.



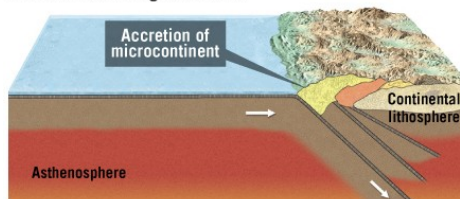
B. The volcanic island arc is sliced off the subducting plate and thrust onto the continent.



C. A new subduction zone forms seaward of the old subduction zone.



D. The accretion of the microcontinent to the continental margin shoves the remnant island arc further inland and grows the continental margin seaward.



11.6 Collisional Mountain Belts

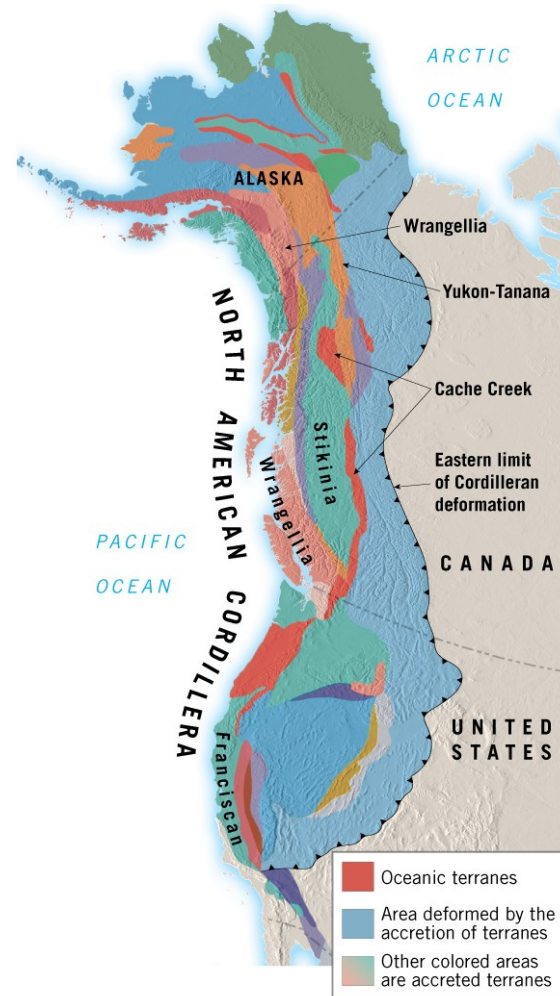
- Cordilleran-type mountain building
 - The nature of terranes
 - Prior to accretion onto the continent, some terranes were microcontinents (similar to Madagascar).
 - Other terranes were island arcs (similar to Japan).
 - Accretion and orogenesis
 - Small features on the ocean floor are subducted with the plate.
 - Large, buoyant features do not subduct, accreted onto the continent.

11.6 Collisional Mountain Belts (cont.)

- Cordilleran-type mountain building
 - The North American Cordillera
 - Many terranes that make up the North American Cordillera were scattered through the eastern Pacific.
 - During the breakup of Pangaea, the Farallon plate began to subduct under North America.
 - Resulted in the piecemeal addition of crustal fragments to the western side of North America

Terranes That Have Been Added to Western North America during the Past 200 Million Years

[Animation](#)



11.6 Collisional Mountain Belts

- Alpine-type mountain building: Continental collisions
 - The zone where two continents collide is called a suture.
 - Typically contains slivers of oceanic lithosphere
 - Most compressional mountains exhibit the deformation of a thick sequence of sedimentary rocks called a fold-and-thrust belt.

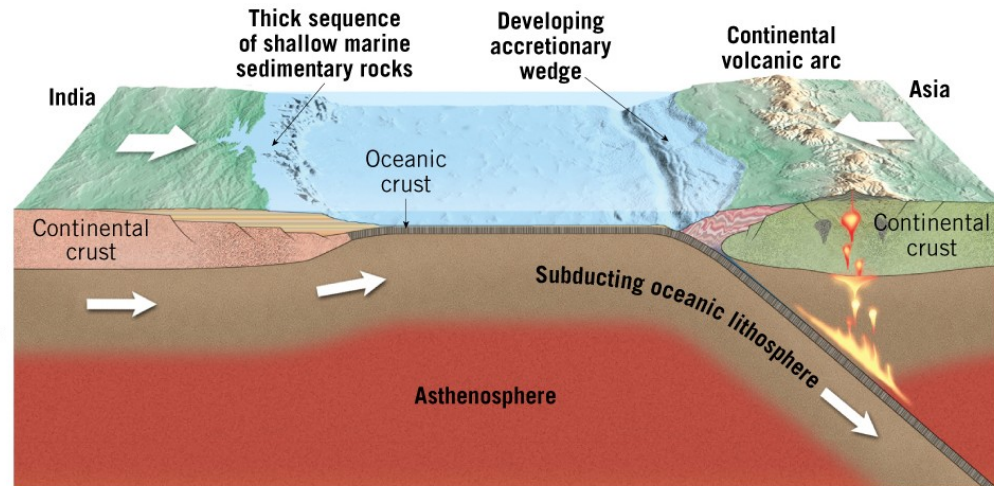
11.6 Collisional Mountain Belts (cont.)

- The Himalayas
 - Collision began about 50 million years ago.
 - India collided with Asia following the subduction of oceanic lithosphere.
 - Precambrian rocks of India resisted deformation, whereas the younger crustal fragments of southeast Asia were highly deformed.

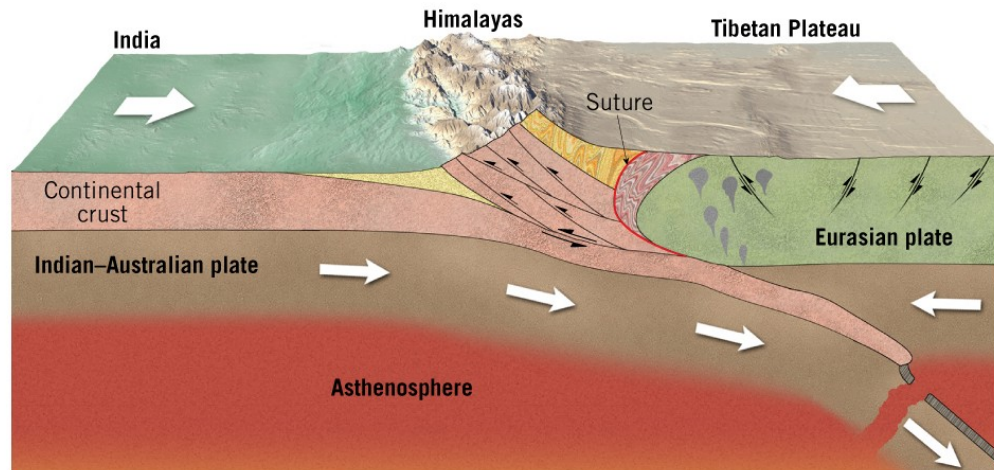
Continental Collision: The Formation of the Himalayas

[Animation](#)

A. Prior to the collision of India and Asia, India's northern margin consisted of a thick platform of continental shelf sediments, whereas Asia's was an active continental margin with a well developed accretionary wedge and volcanic arc.



B. The continental collision folded and faulted crustal rocks along the margins of these continents to form the Himalayas. This event was followed by the gradual uplift of the Tibetan Plateau as the subcontinent of India was shoved under Asia.



11.6 Collisional Mountain Belts

- The Appalachians
 - Of a similar origin to the mountains in the British Isles, Scandinavia, northwest Africa, and Greenland
 - Formed from three main orogenic events that cumulated with the formation of Pangaea

11.6 Collisional Mountain Belts (cont.)

- The Appalachians Taconic Orogeny
 - Volcanic arc located east of North America was thrust over the continental block 450 million years ago.
 - The volcanic rocks and marine sedimentary rocks were metamorphosed and are exposed in New York.

11.6 Collisional Mountain Belts (cont.)

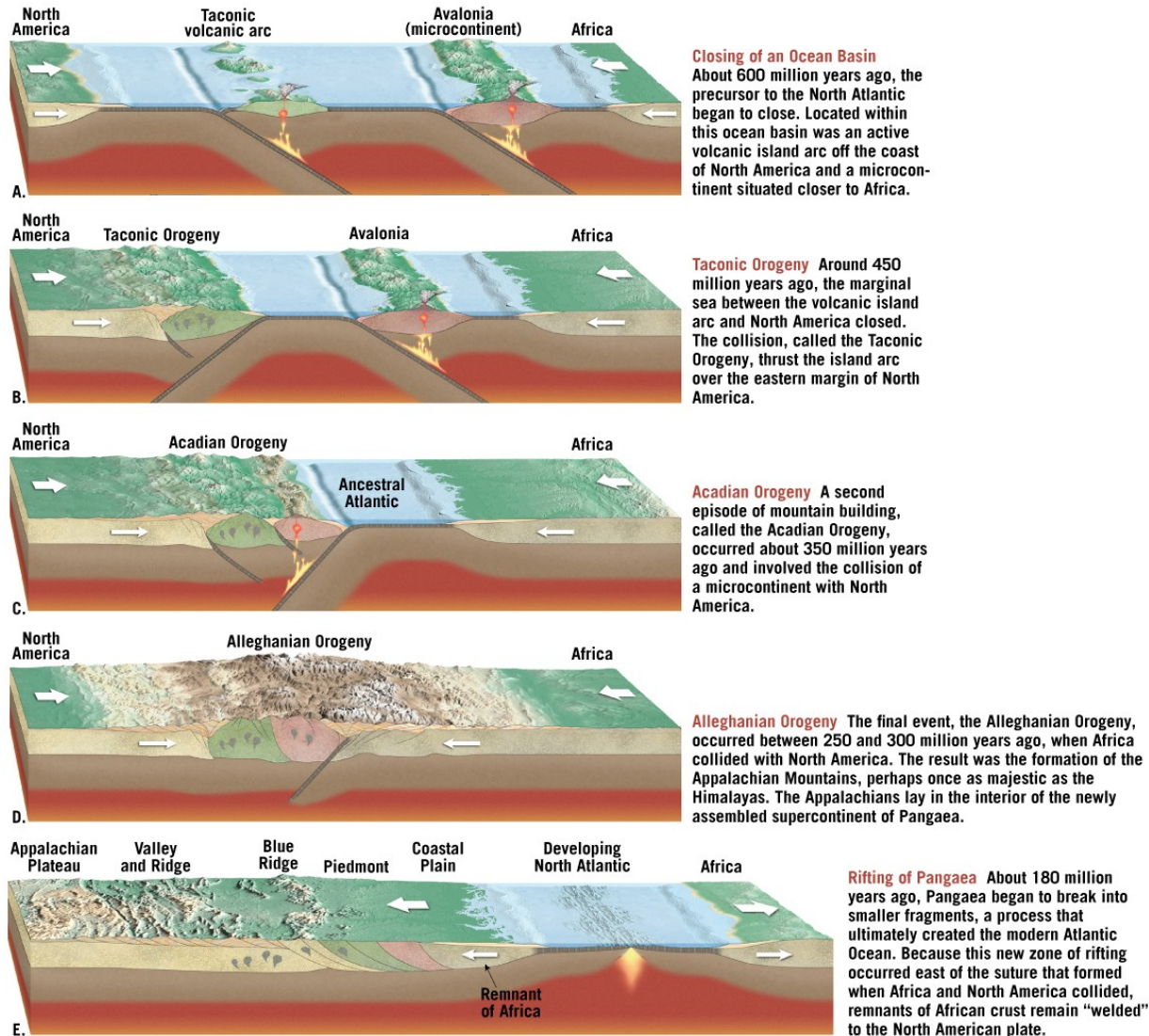
- The Appalachians Acadian Orogeny
 - Continued closing of the ocean basin resulted in a microcontinent colliding with North America 350 million years ago.
 - Thrust faults, metamorphism, and granite intrusions are associated with this event.
 - Substantially added to the width of North America

11.6 Collisional Mountain Belts (cont.)

- The Appalachians Alleghanian Orogeny
 - Africa collided with North America 250–300 million years ago.
 - Material was displaced 250 kilometers inland on North America.
 - Pangaea began rifting 180 million years ago.
 - Rift was eastward of the suture, leaving a remnant of Africa welded to North America.

Formation of the Appalachian Mountains

Tutorial

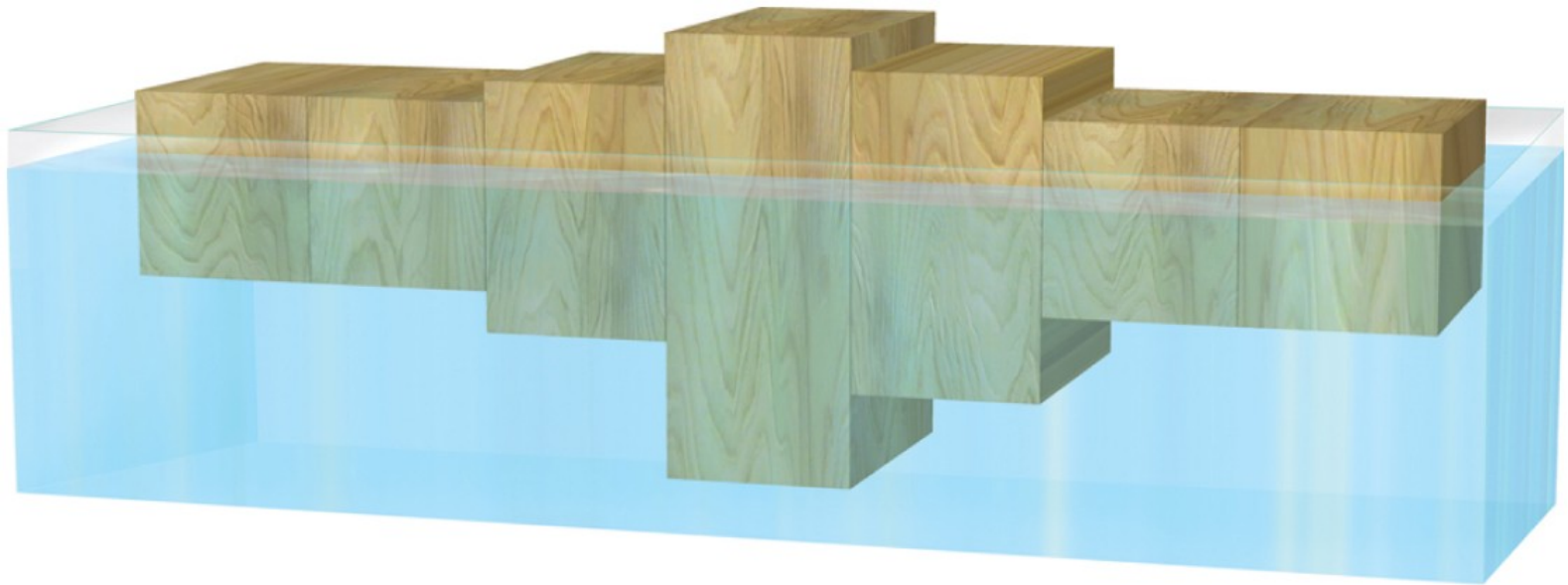


11.7 Vertical Motions of the Crust

- The principle of isostasy
 - Less dense crust floats on top of the denser rocks of the mantle.
 - Isostasy is the concept of floating crust in gravitational balance.
 - Envision a series of different-sized floating blocks on water.

The Principle of Isostasy

[Animation](#)



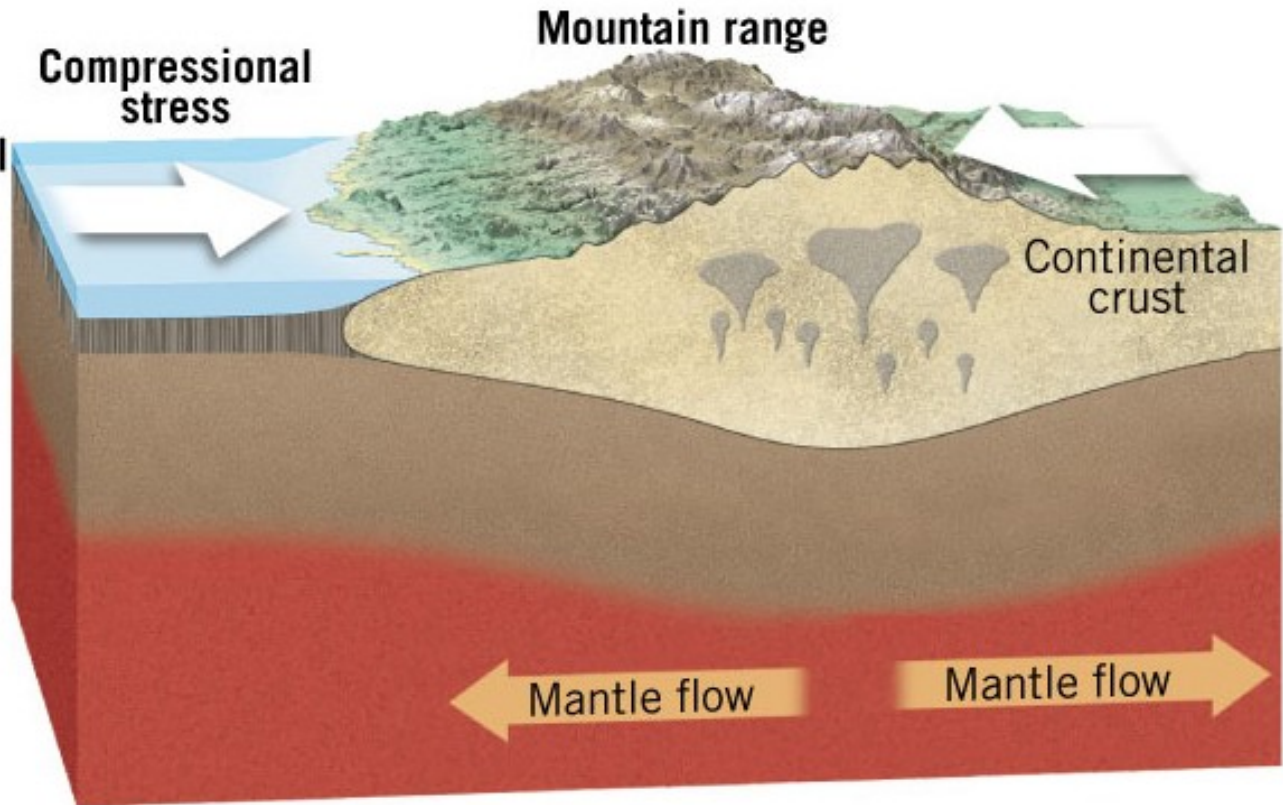
11.7 Vertical Motions of the Crust

- The principle of isostasy
 - How is isostasy related to changes in elevation?
 - If weight is added or removed from the crust, isostatic adjustment will take place as the crust subsides or rebounds.
 - Crustal rebound is present in Canada's Hudson Bay region following the melting of ice sheets in that region.

Effect of Isostatic Adjustment

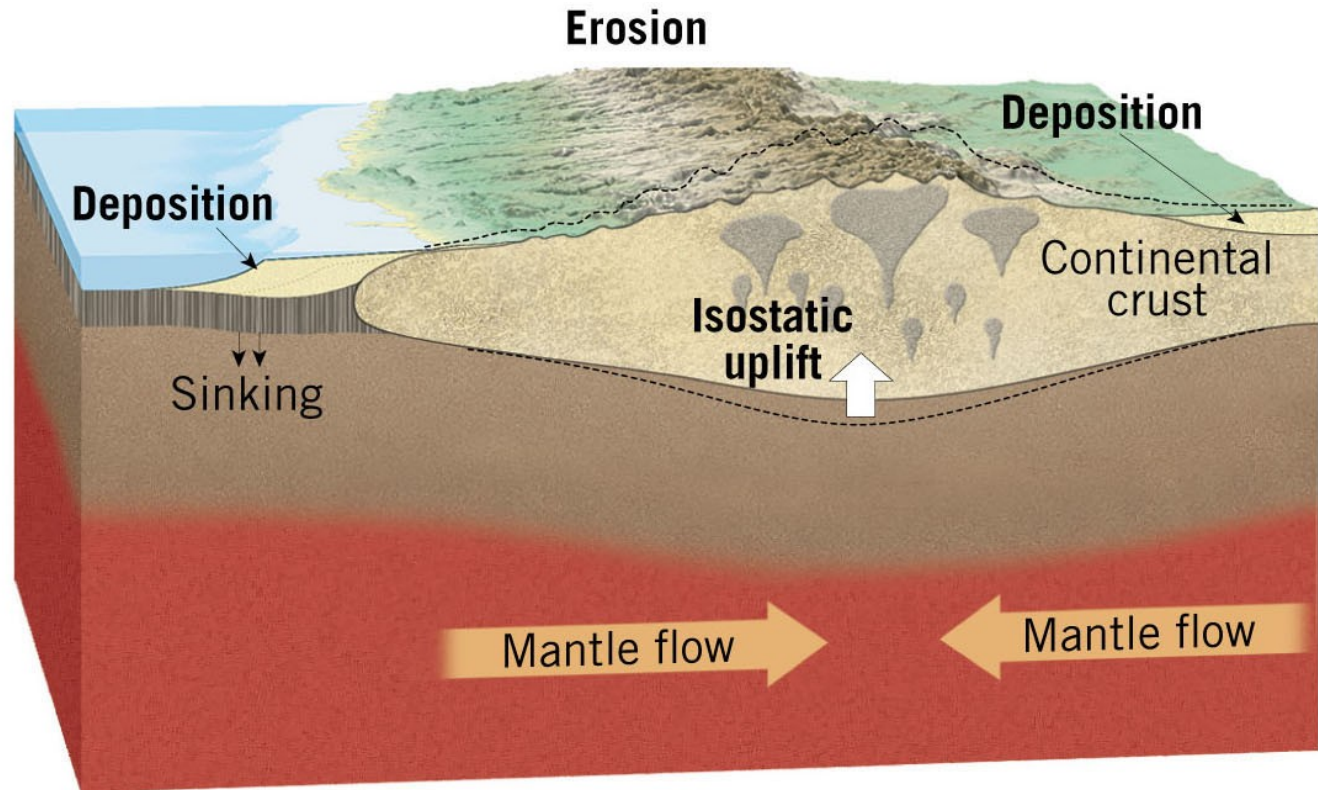
[Tutorial](#)

When compressional mountains are young they are composed of thick, low density crustal rocks that float on the denser mantle below.



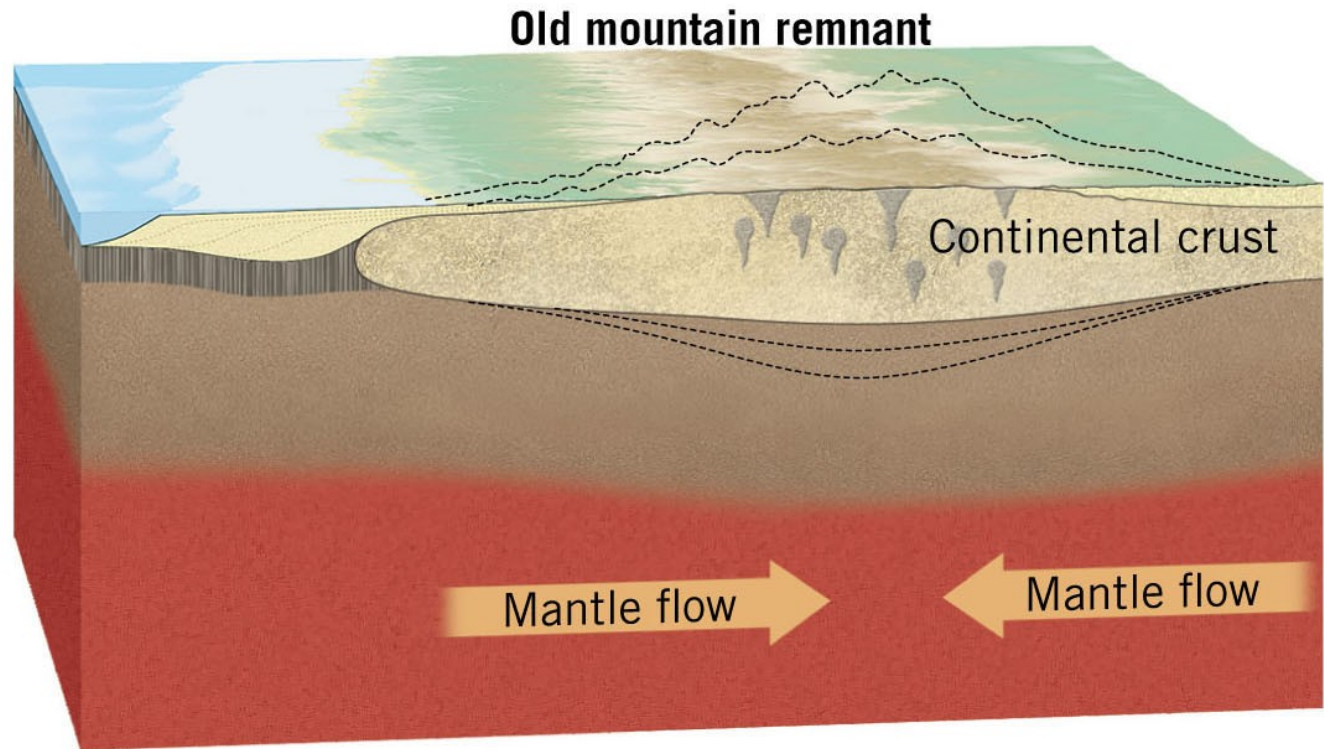
Effect of Isostatic Adjustment (cont.)

As erosion lowers the mountains, the crust rises in response to the reduced load in order to maintain isostatic balance.



Effect of Isostatic Adjustment (cont.)

Erosion and uplift continue until the mountains reach “normal” crustal thickness.



11.7 Vertical Motions of the Crust

- The principle of isostasy
 - How high is too high?
 - As mountains grow, gravity acts on the warm and weak rocks inside the mountains.
 - Eventually, the gravitational forces are so large that the rocks will flow laterally.
 - This ductile spreading and consequential subsidence is called gravitational collapse.

Gravitational Collapse

